Measuring the Reset Distribution: A Tolerance Radius, Referenced to Open-Loop Replay

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***Abstract*—Two choices beneath a robot policy are rarely reported: where an evaluation places the object at reset relative to the demonstrations, and how a fixed budget of demonstrations is distributed. On a low-cost arm with one imitation-learning method (ACT), we place every reset within 2 mm of a target (median error 0.3 mm), log it, and report success against the distance to the nearest demonstration, summarised by a *tolerance radius*: the distance at which success is halfway between its levels near and far from the demonstrations. An open-loop replay of demonstrations has a radius of 16.5 mm (95 % interval 13.0 to 21.5); resets placed by eye from a reference image miss by a median of 15.9 mm. The policy succeeds at larger distances than the replay, but its success depends on location as well as distance (28 of 36 trials inside the convex hull of its demonstrations, 9 of 24 outside), and its fitted radius of 39.0 mm is not established under a pre-specified robustness check. A policy trained on 55 distinct start positions succeeded at 22 of 26 shared test placements, and one trained on 5 demonstrations at each of 11 locations at 16; the paired test does not establish a difference ($p = 0.11$), and the spread layout's own radius was not measured. We propose five reporting-checklist entries.**

I. Introduction

Two choices beneath a robot policy are rarely reported: where an evaluation places the object at reset relative to the demonstrations, and how a fixed budget of demonstrations is distributed. We study both through success as a function of the distance to the nearest demonstration, summarised by a *tolerance radius*: the distance at which success is halfway between its level near a demonstration (the plateau) and its level far from every demonstration (the floor). As a reference, we measure the same radius for an *open-loop replay* (a demonstration's recorded actions executed with no policy or camera): a baseline for the same hardware and control stack. The policy, Action Chunking with Transformers (ACT) [1], is held fixed and is not the object of study. For the replay we find a well-defined radius, close to the median error of resets placed by eye. The policy succeeds at larger distances, but its success depends on location as well as distance, and its fitted radius is not established. Two layouts tested at 26 paired placements are not separated. We conclude with checklist entries.

Imitation policies are known to succeed near demonstrated placements and to fail a few centimetres away [2]–[5], so a reported success rate depends on where resets are placed and how accurately: the *reset distribution*. However, we are not aware of a paper that reports a measured accuracy for its object resets; a benchmark on the arm we use did not record positions, which "prevents analysis of spatial success patterns" [6].

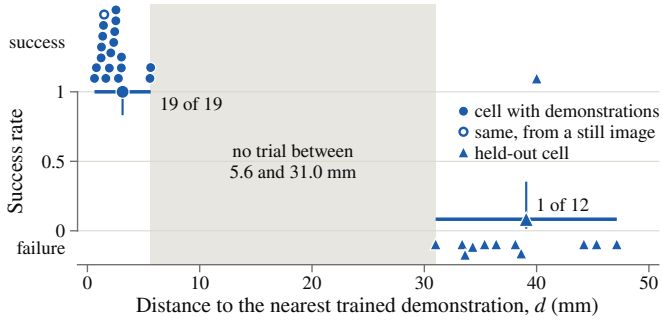


Fig. 1. Our earlier evaluation of the same task, re-analysed. Each mark is one of 31 trials (12 scenes, each run two or three times): its outcome against the distance from the cube's start position to the nearest demonstration the policy was trained on. Large marks: each group's success rate, with a horizontal line over its range of distances and a vertical line for its Wilson 95 % interval. A held-out cell is a grid cell without demonstrations. Start positions were not logged; we measured them afterwards from the trial recordings (open circle: the one trial measured from a still image, not from its video). No trial started between 5.6 and 31 mm.

Our earlier evaluation of the same task illustrates the problem (Fig. 1, Appendix D): of 31 trials whose starts we measured only afterwards, the 19 in cells with demonstrations had started within 5.6 mm of a trained demonstration and all succeeded, the 12 in held-out cells 31 to 47 mm away and one succeeded. No intermediate distance was tested: both rates were determined by where the cube was placed, and neither indicates the distance at which the policy begins to fail.

On a minimal task (one cube, one basket, a low-cost open-hardware arm [7]), each reset is placed by camera overlay within 2 mm of a target, at an adaptively chosen distance, and logged; a fitted success curve gives the radius. The second choice is often made by following a default: the LeRobot guide suggests "at least 50 episodes, with 10 episodes per location" [8]. We compare two layouts of 55 demonstrations: repeated starts at a few locations, or 55 distinct start positions. Both policies are tested at the same placements and compared with the success predicted from the fitted curve.

II. Related Work

Displacement. On real arms, policies have been tested at displacements no finer than about 25 mm [4], at shifts pooled into one rate [3], at placements 150 mm apart [2] or without stated distances [5], [9]. In simulation, DemoGen concludes that "the effective range can be approximated by the union

of the areas surrounding the demonstrated object placements" [2]; we test that hypothesis with a prediction from a fitted curve (Section III).

Evaluation. Absolute success rates "can easily be shifted up or down based on how difficult we make the task" [10]. Resets are made by hand on marks [4], by image overlay [10]–[12], on a grid board [13] or with a guiding tool [14].

Thresholds. Up-and-down staircases [15], [16], Bayesian adaptive placement [17], [18] and psychometric functions fitted with a lapse term [19] are standard in psychophysics [20], [21].

Layout. The documented default repeats locations [8], while a data-scaling study advises a position range "as extensive as possible" [22]. In simulation, 90 uniformly sampled demonstrations gave 66 % success over the workspace, against 29.5 % for 9 points with 10 each [23]. On the same arm, a study with a frozen vision encoder compared one, two and ten positions at 50 demonstrations [4].

Gap. We are not aware of a real-robot study that reports success against the measured distance to the nearest demonstration, estimates a displacement threshold for a manipulation policy with a staircase or a fitted psychometric function, or compares a clustered with a distinct-position layout at a fixed number of demonstrations. Like a recent study of controller gains [24], we measure such choices under controlled conditions.

III. Method

A. Rig and camera measurement

The rig is an SO-101 follower arm [7], front and wrist cameras in a light box, and LeRobot [25], [26] (settings in Appendix A). The task: pick up a 30 mm cube and place it in a basket. A grid of 5×3 cells of 50 mm is fixed to the table by two fiducial tags and is not drawn on it, so the policy sees no marks of position; no demonstration starts in its 4 *held-out* cells.

A *start position* is the cube's position as measured by the front camera, in millimetres in the table's frame. The same detector measures demonstrations and trials, each through a camera-to-table conversion solved from the two floor tags, so an offset common to both cancels in a distance; a printed millimetre scale checks the scale error that remains (Section IV-A). The independent variable is d: the distance from a trial's start position to the nearest one among the demonstrations its policy was trained on.

B. One budget, two layouts

Each dataset holds 55 demonstrations, 5 in each of the 11 other, *trained*, cells (Fig. 2). In the *clustered* layout the 5 demonstrations of a cell start at its centre. This is the structure that the LeRobot guide suggests [8], with fewer repeats per location. In the *spread* layout the demonstrations start at 55 distinct positions, at least 12 mm apart. One operator recorded both sets in four sessions on one day, following one procedure (Appendix A). We trained one ACT policy [1] on each set with identical settings and evaluated both at step 80,000, fixed in advance.

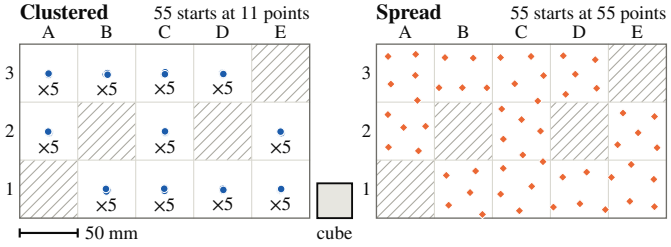


Fig. 2. The two layouts of one budget: measured start positions of the 55 demonstrations of the clustered set and of the spread set. Hatched cells are held out.

C. Trials

In a trial, as before each demonstration, an overlay guides the cube onto a planned target (Fig. 4); the placement is accepted within 2 mm and its measured position logged. The *condition* that then runs is the clustered policy, the spread policy, or a *replay*: a demonstration's recorded actions sent through the same control loop and safety gates, with no policy or camera. The operator scores each trial: only a cube released in the basket counts as a success. We replay 11 clustered demonstrations, one per trained cell, with the arm starting each replay from approximately that demonstration's initial pose; for a replay, d is measured from the start position of the replayed demonstration. For a requested distance L, the target is drawn uniformly among the admissible points whose distance to the nearest demonstration is L. One-up-one-down staircases [15], [16] choose L, larger after a success and smaller after a failure, so that trials concentrate where the outcome is uncertain. A condition's *threshold block* consists of two such staircases, one starting at a small distance and one at a large distance, and *anchor trials* at fixed distances: at $d = 0$ and, for the clustered policy, at 40 mm.

The clustered policy's block and the replay's ran interleaved in one shuffled order, over four consecutive sessions in about 91 minutes, so that both were exposed to the same state of the rig. The *workspace test* followed: 26 placements drawn uniformly, 13 over trained and 13 over held-out cells, each tested with both policies in immediate succession. The spread policy's own block was planned last and was not run. Each trial's condition was concealed from the operator by a code.

D. Analysis

The protocol, with its predictions and this analysis, was frozen before the first analysed trial (Appendix B). For each condition we fit

$$P(\text{success} \mid d) = \gamma + \frac{1 - \lambda - \gamma}{1 + \exp\big((d - r_{50})/w\big)}, \qquad (1)$$

with plateau $1 - \lambda$, the success level near a demonstration, where the lapse rate λ is the share of failures unrelated to distance [19]; floor γ, the probability of success far from every demonstration; width w of the transition; and tolerance radius

r_{50}, where success is halfway between plateau and floor. We report the posterior median of r_{50} with its 95 % posterior interval, or only a bound ("at most" or "at least") where the data do not bound the interval on one side. A *block fit* uses only a condition's own threshold block; all comparisons and predictions use block fits. A *pooled fit* also includes the policy's workspace-test trials and is reported as a check. As a *robustness check*, every fit is repeated with lapse rates up to 0.5 in place of 0.1; if the radius of either fit changes by more than 5 mm, we report that condition's radius as not established. We call one radius *larger* than another when the posterior probability that it exceeds the other is at least 0.95, and *smaller* when that probability is at most 0.05; otherwise the two are not distinguished. The *lookup prediction* for a policy at a placement is the clustered policy's fitted curve at the placement's distance to that policy's nearest demonstration.

IV. Results

A. The camera measurement and the resets

Against the printed scale, a measured 10 mm step deviated by 0.3 mm on average and by at most 0.8 mm. Measured distances were 1.019 times the true distances (1.012 to 1.025 across runs) and are reported uncorrected (Appendix A). Over the 108 scored trials, accepted placements were a median of 0.3 mm from their targets (maximum 0.6 mm), as read by the camera measurement that guided them. Placed by eye, using only a reference image and no overlay or readout, the cube was a median of 15.9 mm from its target (95th percentile 31.0 mm; illustrative, $n = 20$). Resets by image overlay [10]–[12] were not measured.

B. Replay and policy against distance

The replay's success falls within a narrow range of distance: it succeeded in 7 of 7 trials within 10 mm of the replayed demonstration's start, in 6 of 11 between 10 and 20 mm, and in 0 of 4 beyond (Table II, Fig. 3). Its tolerance radius is 16.5 mm (95 % interval 13.0 to 21.5), with a fitted width w of 2.0 mm; the robustness check changes the radius by 0.5 mm.

The clustered policy's success does not decrease steadily with distance: over all its scored trials it succeeded in 11 of 11 within 10 mm of a demonstration, in 6 of 10 between 10 and 20 mm, and in 9 of 12 between 30 and 40 mm, all of the last in held-out cells. The rise at 30 to 40 mm is within sampling variation, but the decline is gradual and location matters besides distance: inside the convex hull of its demonstrations the policy succeeded in 28 of 36 trials and outside it in 9 of 24 (a descriptive split fixed in advance), and in the two interior held-out cells in 13 of 18 trials against 3 of 12 in the two corner ones (an exploratory split). The block fit gives a radius of 39.0 mm (23.0 to 52.0) with a width of 13.7 mm; the pooled fit, which adds the policy's workspace-test trials, gives 36.0 mm (25.0 to 48.5). In the robustness check the block-fit radius changes by 3.5 mm, within our limit of 5 mm, but the pooled-fit radius changes by 6.0 mm, beyond it. By our pre-specified rule, which applies to either fit, the policy's radius is therefore *not established*, and every later result that uses the fitted curve is conditional on it (Appendix B, Table VI).

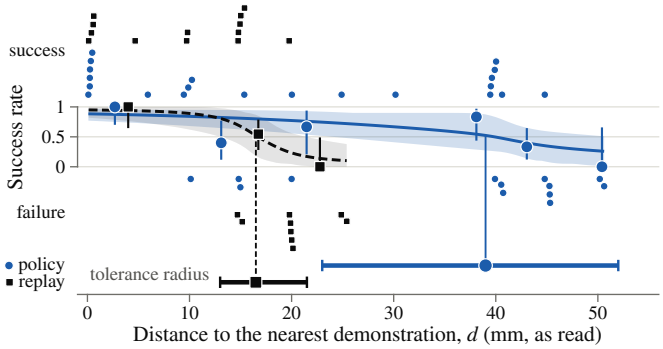


Fig. 3. Success against distance d for the clustered policy and the open-loop replay (threshold blocks; distances as measured, uncorrected). Small marks: single trials, in the row of their outcome. Large marks: success rates in 10 mm bins with Wilson intervals. Curves: posterior means of Eq. (1) with 95 % bands. Below: each r_{50} with its interval. Table II adds the workspace-test trials, so its counts differ.

TABLE I
Tolerance radius by condition, with the farthest success and the nearest failure of each threshold block (mm). First row of a condition: block fit; *pooled*: with the policy's workspace-test trials. The clustered policy's radius is not established (Table VI).

Condition	Scored (void)	Success at $d = 0$	Farthest success	Nearest failure	r_{50} (95 % interval)
Replay	22 (0)	4/4	19.8	14.7	16.5 (13.0–21.5)
Clustered	34 (0)	6/6	44.8	10.1	39.0 (23.0–52.0)
pooled	60 (0)				36.0 (25.0–48.5)
Spread, pooled	26 (0)				at least 19.5

The ordering of the two conditions is the same in each analysis. Under the pre-specified rule the policy's radius is the larger (posterior median of the difference 22.0 mm, 5.5 to 35.5; probability > 0.99). In the robustness check the two intervals still do not overlap (replay 13.0 to 23.5 mm, policy at least 26.0 mm). Without a model (Table II), within 20 mm the two are alike (policy 17 of 21 trials, replay 13 of 18), and beyond 20 mm the policy succeeded in 20 of 39 trials and the replay in 0 of 4. In each of these analyses the policy succeeds at larger displacements than open-loop execution of its own demonstrations on the same hardware, although its nearest failure (10.1 mm) is nearer than the replay's (14.7 mm).

Resets by eye. The median by-eye error, 15.9 mm, lies between 10 and 20 mm, where the clustered policy succeeded in 6 of 10 trials and the replay in 6 of 11, against 11 of 11 and 7 of 7 within 10 mm (Table II).

C. Two layouts at the same placements

In the workspace test the spread policy succeeded at 22 of 26 placements (Wilson interval [27] 66 to 94 %) and the clustered policy at 16 (43 to 78 %); the outcomes differed at 10 placements, 8 in favour of the spread policy and 2 of the clustered policy (exact McNemar test, two-sided, $p = 0.11$; Fig. 5 in Appendix C). The difference is in the predicted

TABLE II

SUCCESSES BY MEASURED DISTANCE d TO THE NEAREST DEMONSTRATION, OVER ALL SCORED TRIALS OF EACH CONDITION. DISTANCES WERE CHOSEN BY STAIRCASES OR FIXED IN ADVANCE, SO THESE ARE NOT RATES OVER A UNIFORM SAMPLE. A DASH: NO TRIAL. BEYOND 25 MM FROM A CLUSTERED DEMONSTRATION, A PLACEMENT LIES IN A CELL CORNER OR A HELD-OUT CELL.

		Clustered policy		Spread policy	
d (mm)	Replay	trained	held-out	trained	held-out
0–10	7/7	11/11	–	11/11	–
10–20	6/11	6/10	–	2/2	7/9
20–30	0/4	4/9	3/4	–	2/3
30–40	–	–	9/12	–	0/1
40–50	–	–	4/11	–	–
50–60	–	–	0/3	–	–

direction, but the test does not establish it: with 26 pairs it detects only large differences (Appendix B), and each layout was trained once, so this compares two trained policies and not two layouts in general. The *lookup prediction*, the clustered policy's block-fit curve applied to each placement's distance to a policy's nearest demonstration, expects 17.7 successes for the clustered policy (95 % predictive interval 10 to 24) and 21.0 for the spread policy (15 to 25). Both observed counts fall within these intervals, which are wide enough that only a large mismatch could have failed: this is weak evidence that distance alone explains the outcomes. Location matters for the spread policy too: inside the convex hull of its demonstrations it succeeded in 21 of 22 trials, and outside it in 1 of 4. The spread policy's threshold block was planned but omitted by the operator's decision, a deviation from the protocol (Appendix C); its workspace-test trials bound its radius only from below (at least 19.5 mm).

Of the four predictions stated before the data, two were met, one was met by its median alone and one was not met (Appendix B); the three deviations from the frozen protocol are listed in Appendix C.

V. PROPOSED CHECKLIST ENTRIES AND DISCUSSION

We propose five entries for the workshop's Design Choice Reporting Checklist, under *demonstrations* and *reset distributions*:

1) **Demonstration start states.** A map of where the object started in each demonstration, how the positions were set, and their scatter.
2) **Reset positions.** Each trial's measured object start and its distance to the nearest demonstration; the success counts by distance; the distances never tested.
3) **Reset accuracy.** The aid used to place the object, the measured placement error, and how the measurement was checked.
4) **Tolerance radius.** Where one curve describes the counts: the distance at which success is halfway between plateau and floor, with an interval or a one-sided bound and a check of the fit. In every case: the farthest success and the nearest failure.
5) **Open-loop reference.** The same for an open-loop replay of demonstrations, with its trials interleaved with the policy's.

The first three require a camera, two printed tags and a log; the last two require the trials of Section IV (34 and 22 trials here, a median of 1.3 minutes each).

Two relationships motivate these entries. First, reset accuracy relative to the distances at which success changes can alter a reported success rate: here, resets placed by eye had a median error of 15.9 mm ($n = 20$, one person), a distance at which the clustered policy succeeded in 6 of 10 trials, against 11 of 11 within 10 mm. Second, demonstration spacing relative to those distances affects how much of a workspace a dataset covers; here the test of the two layouts was inconclusive.

The procedure can be applied to another rig. Two further records concern defaults: the arm's software writes a position gain of 16 where, by its own source, the servo's default is 32 (Table III), and three data-pipeline faults of our earlier work pass the checks that were in place (Appendix E). The overlay, the planner, the trial logs, the measured start states and the frozen protocol are public at https://github.com/daniel-carrera/so101-tolerance-radius-study; the datasets are available on request.

VI. LIMITATIONS

This study has one rig, task, object and policy family, one budget of 55 demonstrations, one training run per layout and 108 trials: a difference between the two policies cannot be attributed to the layouts, and every number is specific to this rig and these runs: the earlier policy of Fig. 1 succeeded in 1 of 12 trials at 31 to 47 mm, the present one in 10 of 20 at the same distances. Beyond half a cell from the clustered demonstrations, distance is not separated from location (Table II). A code concealed each trial's condition, but the scorer also placed the cube and can often recognise a replay from its motion; rescoring 24 trials from video with identities concealed, the same person agreed in 24, a check of consistency and not an independent score. The cube's orientation was set by hand, unmeasured; the arm's start pose was logged, not controlled.

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Appendix A
The rig's settings

Tables III and IV list the rig's settings in the categories of the workshop's checklist. None of these settings was varied, so this study cannot determine how the tolerance radius depends on them.

The table frame: tags, grid and registration. The grid is virtual: nothing is drawn on the floor, so that the policy sees no marks of position. The cameras see only the floor, two fiducial tags and the basket, and the grid appears in the overlay alone (Fig. 4). Two AprilTags (Table III) are taped to the floor outside the grid, one in front on the right of the arm and one at the rear on the left, and tie the grid to the floor. Their positions in the grid's frame were measured once, in a *registration*: a printed sheet carrying the grid's outline and six 36 mm tags was laid where the grid should be, the conversion from pixels to table millimetres was fitted through the six tags over 40 frames, the corners of the two floor tags were stored in that frame, and the sheet was removed. Two findings from the first evening shaped this design. First, a frame solved from the two floor tags alone is fragile: one pixel on one tag corner moves the far side of the grid by up to 2.5 mm, so the two tags are used to carry a frame that six tags established, and not to establish it. Second, the tag detector's corners lie about 0.6 pixels inside a tag's outline at this view (about one pixel per millimetre) and move with the image's brightness, so every tag is measured by lines fitted to its outline instead. Because the camera hangs from the light box's frame, which the box's walls move (closing one zipper turned the camera by 0.1 degrees), each session solves the conversion again from the two floor tags as it finds them.

How the frame performed. At the registration, the six sheet tags fitted one plane to 0.41 mm on average (0.88 mm at most); a tag left out of the fit was predicted by the other five to 0.61 mm on average (1.91 mm at most); and the sheet's printed grid lines, which no fit uses, lay 0.28 mm (root mean square) from the fitted frame. In use, the checks reported in Section IV-A and Table IV apply: steps of 10 mm were read to 0.3 mm on average, repeated readings of one position stayed within 0.3 mm, and accepted placements were a median of 0.3 mm from their targets. The weak point was the camera's

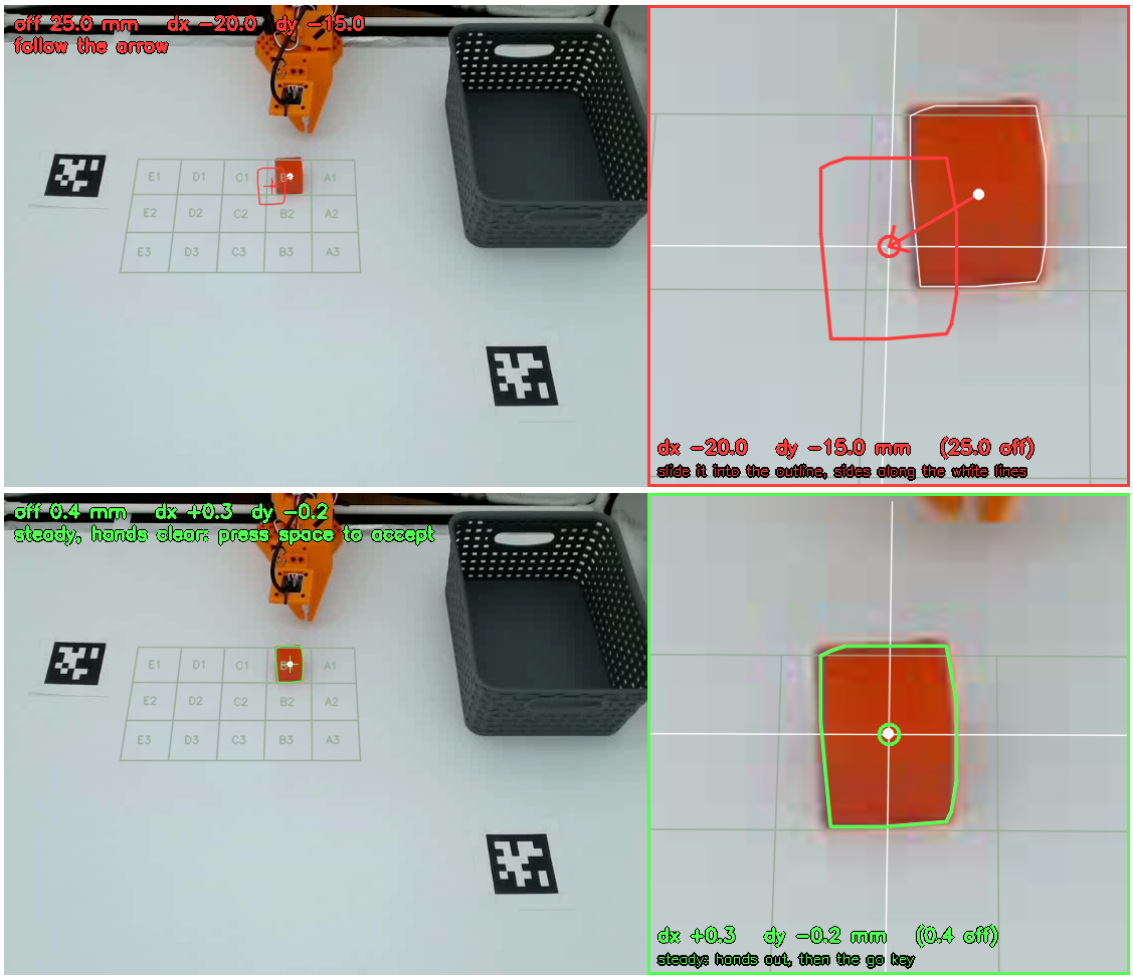


Fig. 4. The placement overlay as the operator sees it, rendered afterwards with the session's own code from the first frame of a counted trial's video. Left: the front camera's image with the grid (which is not marked on the floor) and its cell names, the two floor tags, the arm at its start pose and the basket. Right: the surroundings of the target, magnified. Top: the cube 25 mm from a target chosen for this illustration; the outline shows where the cube belongs, and the readout gives the offset in table millimetres. Bottom: the same frame against the trial's own target, within tolerance (0.4 mm from it).

mounting, not the tags: at the start of the six sessions with counted trials the camera's image stood 0.4 to 1.7 mm from the stored calibration, the two floor tags read 1.6 to 1.8 mm against each other relative to the registration (the rest of the image had moved with them, which indicates a rotation of the camera and not a tag's movement), and two sessions ended on the rule that allows no attempt once the image has moved more than 1 mm within a session. A camera mount that is independent of the enclosure would remove that cost.

The instrument. The front camera detects the cube and converts its position to table millimetres. Each session solves this conversion from the outlines of the two floor tags, averaged over 20 frames, and measures through it; the start position of a demonstration is measured through the tags as that episode's first frame shows them. A calibration stored once serves as a reference only: a session reports a difference above 1.5 mm from it and stops above 5 mm. The detector reports the centre of the cube's visible outline (its top and lit sides) mapped through the table plane, not the centre of its footprint. The same holds for every demonstration and every trial, so the offset cancels in a distance, but only up to a scale factor, which geometry places a few percent above one. Over its 4 runs the scale check gave a factor, each run's measured length divided by its true length (50 to 150 mm), between 1.012 and 1.025 (mean 1.019). The rule fixed in advance: if the runs agree within one percentage point, every distance in the analysis is divided by the mean factor; otherwise, distances are left as measured and the range of the factor is reported as the instrument's accuracy. A placement (Fig. 4) is accepted when the cube is steady within 2 mm of its target and the operator, with hands out of the camera's view, presses a key; only then can the arm move, and the position at that moment is the measured start. For repeatability, the cube was pushed 11 times against a fixed stop and lifted away between readings; the readings stayed within 0.3 mm of their mean. No overlay was involved, so this figure describes the detector alone.

Resets by eye. For each of 20 targets, the cube was returned to its position using a reference image shown beside the live

view, with no overlay and no readout. Number of people who placed the cube by eye: 1; the largest error was 35.5 mm. This corresponds to an evaluation that resets the object "to the same place" from an image or from memory. It does not correspond to an evaluation that overlays a reference image on the live view; we did not measure that case and expect its error to lie between that of resets by eye and that of the resets of this study.

Recording. One operator recorded both sets on one day with the same task text, time limit, cube orientation (square to the grid), start pose, approach side and grasp, in rounds of one episode per trained cell in shuffled order; an episode ended when the operator pressed a key, so episodes differ in length (Table IV). An episode was recorded again when the operator made a handling error, the cube was outside tolerance or a check failed. The protocol plans a practice block that is recorded and discarded; the procedure of the recording day asks for practice without recording, and the logs hold no practice episode. The clustered layout repeats each location to within about a millimetre: the 5 start positions at a location are within 1.4 mm of their mean. In our earlier dataset, where the cube was placed using a grid drawn over the camera image and the start positions were not measured, the two farthest of a cell's five demonstrations were 4.8 to 13.7 mm apart, so a location repeated without a measuring aid is itself somewhat spread. The two layouts compared here are therefore the two extremes of a range; a recording made without a placement aid lies between them. Before training, each dataset was checked for its episode and frame counts, the decoding of both camera streams, frame pacing and torn frames (Appendix E reports the results of the last two), and by the start-state audit: every episode in its planned cell and within 3 mm of its target.

Appendix B
The protocol: predictions, analysis and stop rules

The protocol (version 1.0) was frozen before the first counted trial. A *counted* trial is one run under the frozen protocol, and only counted trials are analysed. Before the freeze, *pilot* trials tested the procedure: a replay pilot, and nine exact repeats with the clustered policy (the second stop rule below). Pilot trials were written to separate logs, and the trial runner does not start a counted trial until the protocol is frozen. The predictions below were written three days before the first counted trial and two before the first pilot trial, and were not changed after the day they were written.

A. Predictions stated before the data

Every radius in this list is a block fit (defined below).

1) *The curve.* The clustered policy succeeds at least 5 times in 6 at $d = 0$ and at most once in 4 at 40 mm, and its radius lies between 5 and 40 mm. *Observed:* 6 of 6, 1 of 4, and 39.0 mm (23.0 to 52.0). Only the median met the prediction: the interval extends to 52.0 mm, and the median of the second fit of the plateau check, 42.5 mm, lies outside the predicted range (Table VI).
2) *Hardware.* The policy's radius is larger than the replay's, by the rule for "larger" below. *Observed:* a difference of 22.0 mm (5.5 to 35.5), and a posterior probability > 0.99 that the policy's radius is the larger. The prediction was met under the pre-specified rule; the policy's radius itself is not established.
3) *Layout.* At the placements of the workspace test, the spread policy succeeds more often than the clustered one. *Observed:* 22 against 16 of 26 ($p = 0.11$). The prediction was not met: the counts are in the predicted direction, but the test does not establish a difference.
4) *Lookup.* Each policy's success in the workspace test matches the lookup prediction: its observed count lies inside the prediction's 95 % interval. *Observed:* 16 against a predicted 17.7 (10 to 24) for the clustered policy, and 22 against 21.0 (15 to 25) for the spread policy. The prediction was met, with wide intervals and conditional on the curve from which it is computed.
5) *Layout and radius.* No prediction; reported as an estimate. *Observed:* not available. The spread policy's threshold block was not run (Appendix C), so it has no block fit and the two radii are not compared; its workspace-test trials alone bound its radius from below, at 19.5 mm.

B. The blocks as planned

Table V gives each block. Staircases move in steps of 10 mm until their first reversal and 5 mm after it, between 0 mm and the block's largest distance; the replay's may request up to 50 mm. A requested distance is met when the target's distance to its nearest demonstration is within 0.5 mm of it, and test placements keep the cube's centre at least 5 mm inside the grid's edge. For a replay the target is drawn uniformly in direction on the circle of that radius around the start of the demonstration played back. The held-out cells are A1, B2, D2 and E3. Inside a trained cell no point is farther than about 35 mm from a clustered demonstration, or than about 17 mm from a spread one; larger distances are tested in held-out cells. The far anchor trials are set at 40 mm and not at 50 mm, where nearly every admissible point lies in a corner cell; at 40 mm the two interior held-out cells also contain admissible points. The workspace test draws 13 of its 26 placements uniformly over the trained cells and 13 over the held-out cells, and tests them in one shuffled order. With placements in trained cells only, the paired test could detect no difference once the radius is large enough for the clusters to cover their own cells; with placements in held-out cells only, it could detect none while the radius is small. A calculation under the lookup model, made before any trial, was used to choose the design. Both policies were to be evaluated at step 80,000; had a run not reached it in time, both would have been evaluated at their largest common saved step, and not below 40,000.

The replay. The replay block plays back 11 demonstrations of the clustered set, one from each trained cell, so that the replay is tested over the same regions of the table as the policy; the planner's seed draws which of a cell's 5 it is, before any

trial. The block's 22 trials replay each demonstration twice, in two different parts of the block: the 4 trials at $d = 0$ are 4 different demonstrations, and each staircase replays 9 different ones. Which demonstration a trial replays does not depend on any outcome. In the replay pilot, each of the 11 demonstrations was replayed once from its own start position: ten succeeded, and the eleventh attempt was terminated by an accidental key press and is logged as a failure. The safety gate that transfers control of the motors allows a replay to begin only when every joint is within 5 degrees of that demonstration's first recorded action, so a replay starts from approximately the pose from which the demonstration started. A policy trial has a looser tolerance of 8 degrees: the policy observes the arm and may start from any nearby pose.

Order and blinding. The trials of the clustered policy's block and of the replay block, 34 and 22, are shuffled into one seeded order, so that the policy and the replay are tested at the same time of day, at the same servo temperature and in the same state of the cameras. The workspace test follows, with the two trials of a pair in immediate succession on the same target and each policy first in half of the pairs; the spread policy's block was planned last and was not run. As run, the interleaved order took four consecutive sessions in about 91 minutes: one session was ended by the rule on camera movement (Table IV) and two after the arm had displaced the basket (see below). The workspace test followed 47 minutes later, in two sessions, the first of which was ended by the same rule; no pair was split between sessions. A trial is identified to the operator by a running number and a code drawn from one shuffled list. Neither indicates which policy runs, which block the trial belongs to, or whether it is a replay.

The second scoring. Every attempt was recorded on video by both cameras. After the last trial, one scored trial in five of each condition of each block (24 of 108) was drawn by a seed and the hash of the trial log itself, so that it could not be known in advance which trials would be scored again; the two trials of a workspace pair were never both drawn. These trials were scored again from the videos alone, under neutral names and in shuffled order, with no condition, distance or first score shown. No second person was available, so the operator who had scored at the rig gave both scores. The two scores agree in 24 of 24 trials; in 0 trials the video did not show the outcome. The score given at the rig remains the trial's outcome regardless of the second score.

Void trials and failures. A trial is void when it could not test its condition: the placement was later found outside tolerance, a camera or motor fault ended the attempt, a person interfered, the safety gate refused the start three times, or a key press ended the attempt within one second of its start. A void trial remains in the log with its reason, and its position in the plan is repeated with the same target, under a new number and a new code. A stop caused by the condition's own motion after the handover (the per-step clamp, the stall rule) is a failure.

The basket and the wrist camera. The basket is fixed to the floor with double-sided tape. At a release, the wrist camera's mount can strike the basket's rim and displace it, and the same contact can shift that camera in its mount. A trial in which this happens keeps its score; the basket is then repositioned with a live overlay of a reference image, and the wrist camera's position is read from the position of the gripper's fixed jaw in its image. The first frames of each trial's own videos were read afterwards. The arm displaced the basket by more than 5 mm in 2 trials (by 35 and 45 mm). In 49 trials the basket began more than 2 mm from its reference position, in all of them between 2.1 and 2.3 mm: one pixel is about 1.1 mm there, so these readings are two pixels from the reference. The number of trials that began with the jaw more than 2 pixels from its position in training was 0 (1.9 pixels at most; the camera's view was offset by about 2 pixels before the first block and was not corrected). The analysis report lists the trials by identifier.

C. Analysis fixed in advance

- *Model and estimate.* Eq. (1), with a posterior on a grid: a flat prior on r_{50} from 0 to 80 mm in steps of 0.5 mm, a flat prior on $\log w$ over 30 values from 0.5 to 20 mm, and $\lambda \in \{0, 0.02, 0.05, 0.10\}$ and $\gamma \in \{0, 0.05, 0.10\}$, the values of each equally weighted. We report the posterior median of r_{50} and its equal-tailed 95 % interval. The staircases only choose where to test.
- *Rationale for the floor.* A trial far from every demonstration can still succeed by chance. In the earlier evaluation one did, through a grasp at the cube's edge at 40 mm, and with the floor fixed at zero that single trial changed the fitted radius from 19.5 to 26 mm. The floor values limit the influence of such a success on the radius. They do not measure how often chance successes occur; the far anchor trials give that as a count.
- *An interval that is open on a side.* The interval is *open* on a side when the posterior at that end of the grid is at least half of what it is at the interval's own bound on that side. We then report "at most" the upper bound (open below) or "at least" the lower bound (open above), and not a radius. This occurs when success is already decreasing at $d = 0$, or is still on its plateau at the largest distance tested.
- *The plateau is checked.* The lapse values above stop at 0.10, which suits a condition that nearly always succeeds at a demonstration's own start. With a lower plateau the model would account for failures near a demonstration by a smaller radius: in simulation, a true plateau of 0.65 reduced the estimated radius by 8 mm. Every fit is therefore made a second time with $\lambda \in \{0, 0.02, 0.05, 0.10, 0.20, 0.35, 0.50\}$, and both fits are reported with the counts at $d = 0$ (Table VI). The first fit gives the radius we report. If the two medians differ by more than 5 mm, we state that the condition's plateau is uncertain and its radius not established, and every result that depends on that radius (a comparison, the lookup prediction, the reset sensitivity, prediction 1) is still computed by its rule and reported with that caveat. Here the clustered policy's pooled fit changed by 6.0 mm, so the caveat applies to this policy; its block

fit changed by 3.5 mm. Over all 60 of its scored trials, the policy succeeded in all 11 trials within 10 mm of a demonstration, in 6 of 10 between 10 and 20 mm, and in 9 of 12 between 30 and 40 mm (all of the latter in held-out cells): the decline is gradual and location matters besides distance (Section IV), which a single curve of distance does not represent.

- *Two fits.* The *block fit* uses a condition's own threshold block only. Every comparison and every prediction uses block fits: the comparison of the policy with the replay, two blocks of one session, and the lookup prediction, which must not have seen the workspace-test trials it predicts. The *pooled fit* adds a policy's workspace-test trials at their measured distances, and is reported alongside the block fit as a check that the two agree. The spread policy's threshold block was not run, so it has a pooled fit only, from its workspace-test trials: placements fixed in advance and not chosen by a staircase, half of them in held-out cells.
- *The radius as the reported quantity.* We report r_{50} and not w: thresholds need far fewer trials than slopes [18], and slopes fitted to adaptive data are biased [28].
- *Without a model.* Success counts in 10 mm bins of d, closed below and open above, with Wilson intervals [27]; the largest d with a success and the smallest with a failure.
- *Comparison of radii.* The posterior probability that one condition's r_{50} exceeds another's, from the two block fits taken as independent, and the difference with its interval. "Larger" means a probability of at least 0.95 and "smaller" one of at most 0.05; anything between is reported as not distinguished by these trials.
- *Workspace test.* The table of paired outcomes, the exact McNemar test, and each policy's success with a Wilson interval. All placements make one test; the counts for trained and for held-out cells apart, a pair counting by the cell its cube was measured in, are descriptive. We call the two policies different at these placements when the test's two-sided p is below 0.05, the level at which we state below what the test can show.
- *Lookup prediction.* For each placement and policy, the predicted probability of success is the clustered policy's block fit at that placement's measured distance to the policy's own nearest demonstration, averaged over the posterior. We report the expected number of successes, the 95 % interval of the predicted count, which carries both the uncertainty of the fit and the chance in 26 trials, the observed count, and whether it lies inside. The fit comes from a block run earlier in the day, so a change in the rig between the two would also appear as a mismatch.
- *Reset sensitivity.* The success the clustered policy's block fit gives at trained positions if resets carry the by-eye errors, beside the same calculation with this study's own placement errors. A calculation, not a measurement. The curve is evaluated at a distance equal to each of the 20 by-eye errors, and to each of the 108 placement errors of the scored trials, and the results are averaged. The result: 79 % with the by-eye errors against 88 % with this study's placement errors, conditional on a fitted curve whose radius is not established; it does not apply to resets made with an overlay.
- *Descriptive only.* Success by placement inside or outside the convex hull of the demonstrations; failure labels by distance; durations; the achieved control rate.
- *Exclusions.* No trial is excluded except void trials and pilot trials, and no outcome is changed after the fact.

D. What the trial budget can show

Before the freeze we simulated each threshold block against success curves assumed true and fitted it as the analysis fits the real trials (300 simulated blocks for each curve). Every figure in this subsection is simulated or computed from stated assumptions; none is a measurement.

One radius. For the clustered policy's block and an assumed curve with $r_{50} = 20.0$ mm and $w = 3.0$ mm, the estimated radius differed from the true radius by -0.2 mm on average, with a root-mean-square error of 1.6 mm; its interval was 7.8 mm wide on average and held the true radius in 97 % of blocks. The same number of trials at fixed, equally spaced distances gave an error of 2.3 mm and an interval 9.0 mm wide, which is why the staircases were kept. The precision depends on how steep the true curve is, which is not known before the trials: Table VII gives each threshold block's simulated error and interval width for a steep curve ($w = 1.0$ mm), an intermediate one ($w = 3.0$ mm) and a shallow one ($w = 10.0$ mm). Where success is already decreasing at $d = 0$, the interval was open below in 55 % of simulated blocks: the estimate is then reported as "at most" and not as a radius.

Two radii. How large a difference between the policy's radius and the replay's radius the rule for "larger" can detect depends on how steep the two curves are, and the rule can also err when the two radii are equal. Table VIII gives, for a replay radius of 15.0 mm and 300 simulated pairs of blocks per case, how often the rule returned "larger". At equal radii it returned "larger" in 2 % of pairs and "smaller" in 3 % (both curves of width 3.0 mm).

Two policies at the same placements. With 26 pairs and true success rates of 40 and 90 %, the exact McNemar test reaches $p < 0.05$ in 96 % of cases, by an exact sum that takes the two outcomes at a placement as independent. It cannot reach that level with fewer than six discordant pairs, and with six only if all six favour the same policy. The test therefore detects only large differences; where the two policies are closer, we can report only the counts and their intervals.

E. Stop rules

Table IX gives the criteria fixed in advance. What happened at each is as follows. *Before recording:* the scale check passed (Section IV-A) and the camera's reference was stored. The rig check passed its five steps on the arm and the cameras and failed its last step, a comparison of the front camera's image with a stored one, on a stream that had opened without the manually set exposure; the image matched the stored one again

a minute later. No complete pass of that check was recorded afterwards, and the rule for a missed line (omit the spread set) was not applied: both sets were recorded the next day. From then on every session, and the recorder at every episode, compared the image's white and black levels with those of the stored calibration before proceeding. This happened two days before the protocol was frozen, so it is reported here and is not among the deviations from the frozen protocol. *After recording:* both datasets passed their audits, and the pilot gave 8 successes in 9 exact repeats (trials at a demonstration's own start), the minimum required. *During the trials:* the two interleaved blocks were complete at 14:54, before the set time of 15:00. *End of the trials:* those two blocks and the workspace test were complete. The spread policy's block was then omitted by a decision of the operator and not under the third criterion; it is the first deviation listed in Appendix C.

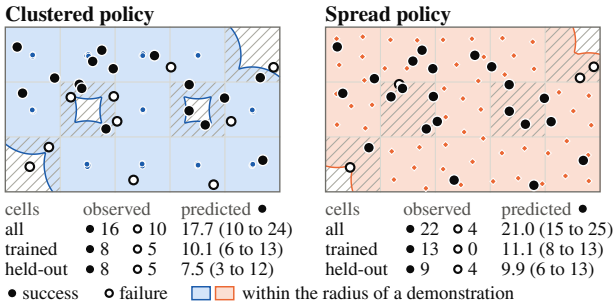


Fig. 5. The workspace test: each policy's demonstrations and the outcome at each of the 26 shared placements. In both panels the discs have the clustered policy's fitted radius r_{50}, which is not established; they are not a measured region of success. Below: successes and failures observed, and the successes the lookup prediction expects with its interval, for all placements and by kind of cell.

Appendix C
Deviations, trial counts and void trials

The decision log records 3 deviations from the frozen protocol. First, the spread policy's threshold block (24 trials) was planned and not run. The protocol's rule for omitting it applies when the two interleaved blocks are not complete by a set time, and they were; the operator omitted it on the day of the trials, to allocate the remaining time to the by-eye resets, the repeatability check and the second scoring. At that point the operator had scored every trial without knowing its condition, and no count by condition had been computed. As a consequence the spread policy has no threshold block, and the two policies' radii are not compared. Second, the repeatability check, which the protocol places before any trial, was run after the last trial on the same day. Third, the check of the camera against the floor tags that the protocol places at the end of each block was not run as a separate step: the start of the next session, about 47 and 39 minutes after the last trial of the two blocks, made the same check, and both passed (0.8 and 0.7 mm from the stored calibration, where a session stops above 5 mm). The second and third do not change what any trial measured. Table X gives the trials planned, scored and void in each block. Void trials: none. Of the descriptive results planned (Appendix B): every failure carries the label "grasp": 23 for the clustered policy (at 10.1 to 50.6 mm), 9 for the replay (14.7 to 25.4 mm) and 4 for the spread policy (16.1 to 38.3 mm). A trial's record lasted a median of 24 s, 24 s and 20 s, which is longer than the attempt itself. Inside the convex hull of its demonstrations the clustered policy succeeded in 28 of 36 trials and outside it in 9 of 24; the spread policy in 21 of 22 and 1 of 4. Number of days on which trials ran: 1. Number of people who placed the cube and scored: 1. Fig. 5 shows the workspace test. In it both policies succeeded at 14 placements and neither at 2; the placements were at most 50.5 mm from a clustered demonstration and 38.3 mm from a spread one. The replay block played back 11 demonstrations of the clustered set (planned: 11).

Appendix D
The earlier evaluation, re-analysed

The earlier evaluation of Fig. 1 tested one ACT policy on the same task, on this arm, with the front camera on a tripod instead of in the light box. Its 31 trials did not log the cube's start position. We measured it afterwards with the detector that had measured that policy's demonstrations: from the first frame of the trial's video in 30 trials, and from a still image saved at the start of the session for the one trial that was not recorded on video. The policy had been trained on 49 of the 55 demonstrations of its dataset, and distances are to the nearest of those 49.

The 19 trials in cells with demonstrations started 0.6 to 5.6 mm from the nearest trained demonstration, 17 of them within 3.1 mm, and all succeeded (Wilson 95 % interval 83 to 100 %). The 12 trials in held-out cells started 31 to 47 mm away and one succeeded, at 40 mm (1.5 to 35 %). With a second conversion from pixels to millimetres, one map for every image, the two groups are at 0.1 to 4.9 mm and at 32 to 47 mm: the untested range does not depend on the conversion. In nine of the trials the cube had been placed on a demonstration's recorded start with an overlay that drew that demonstration's cube over the live view and showed the offset in pixels; those placements were 0.8 to 3.0 mm from it.

Three factors limit what that evaluation can show, in addition to the untested range. Every near trial was in a cell with demonstrations and every far trial in a held-out cell, so it cannot separate distance from cell. The 31 trials are 12 scenes (a cell, or one demonstration's start re-created), each run two or three times, so they are not 31 independent tests and the intervals above are too narrow. Finally, the distances include a scale error that was not measured: the detector reads the centre of the cube's visible outline, not of its footprint, which, by geometry, makes measured distances a few percent too long. The present study begins by measuring that scale.

Appendix E
Faults in the data pipeline of the earlier work

The earlier project's own records show three faults, each of which passes a check that the project had and each of which the project itself found within a day (Table XI). We report them because each lies beneath the policy, in the data it is trained on or in how it is executed, and none would normally appear in a paper's method section. We did not measure their effect on the policy. None of three public dataset checkers lists, in its own description, a check of the decoded image for torn frames [29]–[31]; this is a statement about what the descriptions list, not a test of the tools. This study's own recording encountered torn frames again, and two further faults of the cameras; its records document them. Two earlier attempts at the clustered set, recorded on the previous day, were not used: 4 episodes were discarded after the front camera's image changed in the middle of an episode, and 27 because the wrist camera had started with a different tone mapping of its image in 21 of them. In the two datasets that were used, a scan written for this rig (the earlier scan classified intact frames as torn on a white floor) flagged 9 frames in 6 of the 110 episodes; on individual inspection, 7 are torn, all in the top rows of the front camera's image, outside the area of the task. All episodes were kept and used for training. No episode failed the pacing check, which here counts the control loop's cycles against wall-clock time rather than the nominal time stamps (Table IV).

TABLE III
THE RIG'S SETTINGS, IN THE CATEGORIES OF THE WORKSHOP'S DESIGN CHOICE REPORTING CHECKLIST (CONTINUED IN TABLE IV).

Category	Setting	Value
Hardware and embodiment	Arm	SO-101 follower and leader [7]; both built by the operator from one seller's SO-101 kit (WowRobo: printed parts, servos and wrist camera), with no change reported beyond the kit's wrist-camera mount; the follower's base is clamped at the front of the light box, on the box's centre line
	Gripper	The kit's standard printed jaws, with nothing added to them; fully open, the jaw tips are 151 mm apart (the operator's ruler)
	Object and basket	Cube of 30 mm: painted wood, orange, one face with rounded edges (placed face down); mass 20 g. Basket: a grey plastic storage basket moulded in a woven pattern, 256×201 mm at the rim by the vendor's listing, 104 mm tall, its base 225×170 mm by the operator's ruler. It stands on double-sided tape beyond the grid's column A, clear of the box's pipes, its nearest corner about 113 mm outside the grid (read from the front camera's image; accurate to about 10 mm); its position is defined by a reference image, and it is returned there with a live overlay whenever the arm has displaced it
	Workspace	Grid of 5×3 cells of 50 mm, fixed to the table by two tags (AprilTag 36h11, ids 0 and 1, printed at a nominal 60 mm (measured: 59.7×60.0 and 59.8×60.0 mm)); light box: the 81 cm (32 inch) cubic photography light box of the VLA-REPLICA cell [12] (Glendan): a frame of plastic pipes, walls lined with reflective silver fabric, a white floor sheet; three LED panels under the top, evenly spaced, above a white diffuser sheet fastened with hook-and-loop tape to the top's edges and the front poles, operated at the second-lowest of the dimmer's brightness steps. The front flap was zipped shut except along the strip where the arm is clamped, and every other opening was closed with the box's own flaps except the side window on the arm's left, which stayed open. The lamps' power and colour temperature were not read
Motors and actuation	Servos	Feetech STS3215; the follower's six are the 12 V variant (labelled STS3215-12V in the seller's pictures; STS3215-C018, geared 1/345, in the kit's listing; the servo's own listings give a stall torque of 30 kg cm at 12 V), on a 12 V, 8 A supply (its label); the leader's six are the kit's 7.4 V variants, geared 1/345, 1/191 and 1/147. The labels of this arm's servos were not read
Kinematics	Joints, calibration	Five arm joints and a gripper; calibrated by the operator with LeRobot's procedure (each joint set by hand to the middle of its range, then moved through its whole range), following the kit seller's video guide. The calibration files used throughout were written on 2026-07-03 and not since, no servo was taken off its horn when the arm was moved into the light box, and every session loaded the files without reporting a mismatch with the servos
Controller gains and rates	Servo position loop	P = 16, I = 0, D = 32, in the servo's register units, on all six servos. LeRobot's follower code writes these values, its defaults, each time it connects, and no session of this study overrides them. LeRobot's source gives the servo's own defaults as P = 32, I = 0 and D = 32; according to that source, only P differs from the servo's default. The values were read from the code that drives the arm and were not read back from the servos
	Control rate	30 Hz requested; 29.1 Hz achieved in policy trials
	Limits on a command	Per-step clamp of 15 degrees; a policy's first command within 8 degrees of the arm's pose on every joint, a replay's within 5; stall rule: an attempt ends when one joint is at least 13.5 degrees from its command (0.9 of the clamp) and no joint has moved 1.5 degrees over the last 30 steps
Action spaces	Action	A position target for each of the six servos at every control step: the five arm joints in degrees from the middle of their calibrated range, the gripper from 0 to 100 over its range. A demonstration's actions are the leader arm's joint positions
Observations	Cameras	Front: Logitech HD Pro Webcam C920, mounted inverted on the light box's top rear pipe, on the box's centre line, its image rotated by 180 degrees in software; held by a printed plate ($123 \times 30 \times 4$ mm) with two snap hooks of the VLA-REPLICA camera mount on the pipe and a 1/4-inch screw with two lock nuts in the camera's tripod thread; lens 674 mm above the floor and 810 mm in a straight line from about the middle of the grid (tape measure), which is a view from about 56 degrees above the floor (derived from the two lengths). Wrist: the kit's 2-megapixel USB colour camera on the kit's printed wrist mount (the computer lists it as USB2.0_CAM1, USB vendor 1443, product 37424; no model name was read from it). Both 640×480 pixels at 30 Hz; front camera set manually in the vendor's software (Logi Tune 3.15.62), which must remain running to maintain these settings: exposure 115 with gain 0, white balance 4155 K, focus 0, brightness 77, contrast, saturation and sharpness 128, anti-flicker 60 Hz, and a digital zoom of 118 (100 is none) with pan -7200 and tilt 25200, all in that software's units. Wrist camera: exposure and white balance automatic (its default); focus set by hand on its lens
	Other inputs	The follower's six joint positions, in the action's units, at the current step only

TABLE IV

THE RIG'S SETTINGS, CONTINUED: DEMONSTRATIONS, THE POLICY, AND THE RESET DISTRIBUTIONS.

Category	Setting	Value
Demonstrations	Interface	Leader arm. Recording rate in Hz: 30 requested; the recording loop held a median of 29.99 in both sets, and 29.96 in its slowest episode. Episode length in seconds: 17.3 to 26.0 (median 20.9 in the clustered set, 20.0 in the spread set); an episode ends when the operator presses a key
	Budget and layout	55 per dataset. Clustered: 5 at each of 11 points. Spread: 55 positions, planned at least 12 mm apart (measured: at least 17.5 mm)
	Recording	4 recording sessions on one day, in the order clustered, spread, spread, clustered; practice before the first session was not recorded; 2 clustered and 1 spread episodes were recorded again. Recorded on the previous day and not used: 31 clustered episodes (Appendix E) and 5 test episodes of the recorder
	Start states	Set with the overlay, then measured from each episode's first frame: 0.7 mm from the planned target at the median, 2.7 mm at most
	Trained on	55 clustered and 55 spread episodes; held out for an evaluation loss: none
Policy (held fixed; not a checklist category)	Model and training	ACT [1], 51.6 M parameters, ImageNet-pretrained ResNet-18 backbone, chunk of 100 steps, batch 32, learning rate 10^{-5}, seed 1000, image augmentation off; one run per dataset on one A100 GPU, LeRobot 0.6.1; evaluated at step 80,000
	Execution	Temporal ensembling with coefficient 0.01 and inference at every step; attempts limited to 45 s; LeRobot at commit 6adf51511b
Simulation		None: every trial ran on the arm
Reward design		None: imitation learning. Outcomes were scored by a person on a graded scale, with success only for a cube released in the basket
Reset distributions	Object	Placed with the overlay within 2 mm of a planned target; accepted placements 0.3 mm from the target at the median, 0.6 mm at most; every measured position and its d logged
	Arm	Set by the operator with the leader arm before every trial; no pose is stored or commanded, and there is neither a physical stop for the follower nor a rest for the leader. In this pose the arm is folded and the gripper is closed. As recorded at the start of each of the 108 counted trials, the elbow ranged over 1.0 degrees and the shoulder's lift over 3.2, the shoulder's pan over 10.7, the wrist's roll over 15.4 and the wrist's flex over 20.0
	Camera	Each session solves the camera-to-table conversion from the two floor tags and measures through it. The result is compared with a stored calibration: a difference above 1.5 mm is reported and one above 5 mm stops the session; in the kept start frames of the six sessions with counted trials it was 0.4 to 1.7 mm. No attempt is made once the image has moved more than 1 mm from the session's own solve: the reading before an attempt was 0.57 mm at the median and 0.99 mm at most

TABLE V
THE BLOCKS AS PLANNED. DISTANCES IN MILLIMETRES.

Block	Trials at $d = 0$	Trials at 40 mm	Two staircases: starts; trials each	Largest distance
Replay	4	–	5 and 25; 9	50
Clustered policy	6	4	10 and 40; 12	50
Spread policy	4	–	5 and 25; 10	50
Workspace test	26 placements, each with both policies: 13 in trained cells, 13 in held-out cells			

TABLE VI
THE PLATEAU CHECK: EACH FIT MADE WITH THE LAPSE VALUES OF THE ANALYSIS (UP TO 0.10) AND AGAIN WITH THE LONGER LIST (UP TO 0.50). POSTERIOR MEDIAN AND 95 % INTERVAL OF r_{50} IN MILLIMETRES; AN INTERVAL THAT THE DATA DO NOT BOUND ABOVE IS GIVEN AS ITS LOWER BOUND, AND THE MEDIAN OF SUCH A FIT IS SHOWN ONLY TO GIVE THE SHIFT, WHICH IS THE SECOND MEDIAN MINUS THE FIRST. THE SECOND COLUMN GIVES THE SUCCESSES AT $d = 0$ AMONG THE FIT'S TRIALS.

Fit	At $d = 0$	$\lambda \leq 0.10$ median	$\lambda \leq 0.10$ interval	$\lambda \leq 0.50$ median	$\lambda \leq 0.50$ interval	Shift
Replay, block	4/4	16.5	13.0–21.5	17.0	13.0–23.5	0.5
Clustered, block	6/6	39.0	23.0–52.0	42.5	≥ 26.0	3.5
Clustered, pooled	6/6	36.0	25.0–48.5	42.0	27.5–58.0	6.0
Spread, pooled	none	28.5	≥ 19.5	29.0	≥ 20.0	0.5

TABLE VII
SIMULATED BEFORE ANY TRIAL: ROOT-MEAN-SQUARE ERROR OF THE ESTIMATED RADIUS / MEAN WIDTH OF ITS 95 % INTERVAL, IN MILLIMETRES, FOR EACH THRESHOLD BLOCK AS PLANNED AND THREE WIDTHS OF THE ASSUMED CURVE. THESE VALUES ARE SIMULATED, NOT MEASURED.

Block	Steep	Intermediate	Shallow
Clustered policy	0.8 / 4.2	1.6 / 8.1	4.2 / 18.4
Replay	1.0 / 6.3	2.1 / 12.6	5.2 / 23.6
Spread policy	0.9 / 5.3	1.9 / 11.3	4.8 / 21.7

TABLE VIII
SIMULATED BEFORE ANY TRIAL: SHARE OF PAIRS OF BLOCKS (CLUSTERED POLICY, REPLAY) IN WHICH THE RULE RETURNED THAT THE POLICY'S RADIUS IS THE LARGER, IN PERCENT, BY THE TRUE DIFFERENCE OF THE TWO RADII AND THE WIDTH OF BOTH CURVES. THESE VALUES ARE SIMULATED, NOT MEASURED.

Width	equal	+3 mm	+5 mm	+8 mm	+10 mm	+15 mm
1 mm	3	56	92	100	100	100
3 mm	2	17	38	72	86	99
6 mm	3	10	20	38	55	83

TABLE IX
STOP RULES FIXED BEFORE THE EXPERIMENT.

When	Pass criterion	If missed
Before recording	Rig check passes; camera baseline stored; scale check within 1.5 mm mean absolute error, no step deviating by more than 3 mm	Repair before recording; if that is not possible in time, omit the spread set
After recording	Both datasets recorded and audited; in a pilot, a clustered checkpoint succeeds in at least 8 of 9 exact repeats	Stop: no submission
During the trials	The clustered policy's block and the replay block complete by a set hour	Skip the spread policy's block; run the workspace test only if all its pairs fit
End of the trials	The clustered policy's block and the replay block complete	Report the completed blocks, and state what was planned and not run

TABLE X
TRIALS PLANNED, SCORED AND VOID, BY BLOCK.

Block	Planned	Scored	Void
Replay	22	22	0
Clustered policy, threshold	34	34	0
Workspace test, both policies	52	52	0
Spread policy, threshold	24	0	0
All	132	108	0

TABLE XI
FAULTS RECORDED IN THE EARLIER PROJECT'S DATA AND TRIALS.

Fault	Extent	Checks that pass despite it	How it was found
Torn video frames: one frame composed of parts of two frames	31 frames flagged in 18 of the 55 episodes behind the earlier evaluation	The dataset check, which decodes the first and last frame of every episode, and ffmpeg's error report: both pass on these files	The project's own scan of every decoded frame for a seam that neither neighbouring frame has, written after torn frames had been noticed visually in its first dataset (171 frames in 46 of its 53 episodes)
Episodes captured more slowly than their time stamps say	16 episodes at about 28 Hz, stamped 30 Hz	No warning at recording, where the shortfall was logged at debug level only; the dataset check of frame pacing, which reads nominal time stamps and still reports that the loop never stalled	Not recorded. Notes from the same evening give those episodes' frame counts, about 700 where 25 s at 30 Hz correspond to 750
An execution setting off without notice: temporal ensembling of the policy's actions	2 trials, kept apart from the reported ones	A unit test of the option's default, which read the value written in the source and not what a run parsed	The operator noted discontinuous motion in both trials; neither session's log contains the line that reports ensembling as enabled